Sequential Domain Adaptation on Heterogeneous Clinical Resources for Biomedical NER

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Abstract

In the field of Natural Language Processing (NLP), there is a high abundance of biomedical and clinical datasets. However, finding a specific dataset tailored to a clinical sub-domain for information extraction, model training, and evaluation remains a significant challenge. This scarcity of domain-specific data hinders the ability of models to learn underlying patterns effectively, limiting their performance and generalization. To address this issue, we propose leveraging sub-domains of clinical resources to enrich pre-trained models with task-specific knowledge through fine-tuning. In particular, our paper focuses on using continuous learning for transfering knowledge from pre-trained models across various sub-domains of biomedical resources. Using this approach, we develop an adaptable named entity recognition (NER) model which extracts and identifies biomedical entities across different resources. The focus of our research concerns with three core areas in biomedical NLP: scientific papers, clinical trials, and patient profiles. We use sequential fine-tuning with layer-freezing to mitigate catastrophic forgetting, ensuring that knowledge from previously learned sub-domains is retained while adapting to new ones. Additionally, we provide empirical validation on three diverse biomedical sub-domains, demonstrating the effectiveness of our approach.

Keywords

Natural Language Processing, Named Entity Recognition, Biomedical Sub-Domains, Domain Adaptation,

1. Introduction

Biomedical text mining has become a crucial component of modern medical research, supporting applications such as literature curation, clinical decision support [1], and large-scale knowledge extraction [2]. Central to this process is named entity recognition (NER), which serves as the foundation of biomedical text mining, for extracting relevant concepts and their classification into labels such as diseases, genes, chemicals and anatomical structure.

However, despite the availability of large annotated datasets and domain-specific models such as BioBERT [3] and PubMedBERT [4], NER remains a challenging task due to linguistic variability, context ambiguity, and overlapping entity boundaries in biomedical text.

Our contribution is part of the larger AVATAR project[1], in which the generation of digital avatars ensures the anonymisation of personal data. The General Data Protection Regulation (GDPR) protects sensitive health data from potential misuse and risk of reidentification by requiring data to be fully anonymous so that the individual is no longer identifiable. To preserve the confidentiality of the data, the goal is to enforce GDPR compliance so as to not reveal the identity of the users while advancing further research in this area.

As a part of this project, we aim to develop a system, through which users like clinical trial designers, medical professionals, and researchers can retrieve anonymised patient datasets. The initial interface of this system utilizes an NER model which carries out information extraction on free-text and PDF data. In particular, our work focuses on extracting relevant clinical concepts from unstructured data which includes both free-text and PDF using pre-trained model which is enriched through continuous learning over different biomedical resources. For the NER task, we use encoder-based models like BioBERT that has been fine-tuned on biomedical research papers, which are further adapted to the BC5CDR dataset

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[1]https://www.avatar-projekt.de/en/

[5], and subsequently to the Chia[6] and MACCROBAT datasets [7] through partial layer freezing. This ensures that our model does not overfit while in the process of fine-tuning to the succeeding datasets and preventing catastrophic forgetting from the preceding ones.

We focus on robustness and model generalization across different clinical domains. Using a generalized and adaptable model, we then develop an interface through which the users can enter clinical information such as patient clinical profiles, clinical trials, scientific publications etc. to retrieve relevant clinical concepts.

2. Background

2.1. Named Entity Recognition in Biomedical Resources

To process the clinical data, we utilise Named Entity Recognition (NER) in the context of biomedical data to identify and classify concepts such as disease, genes, drugs, and other relevant entities from unstructured resources. A major challege in biomedical NER is caused by the diverse terminology which includes a quite specialized vocabulary and context-dependent meanings of terms. The varying annotation schemes such as BIO (Begin-Inside-Outside), BIOES (Begin-Inside- Outside-End-Single) and nested BIO formats along with nested structures, and overlapping annotations further complicate standard approaches in NER. For example, a single clinical text may contain both a disease mention ("diabetes") as well as its subtype ("type 2 diabetes mellitus") requiring an accurate distinction between them [8]. Recent advances in NER systems, especially in the biomedical domain, have led to the development of domain-specific architectures and other specialized approaches to handle such challenges posed by complex data through the inclusion of knowledge graphs [9] and using attention mechanism to better capture dependencies across longer biomedical texts[3].

For enriching models with biomedical information, fine-tuning allows pre-trained models to adapt to specific tasks or domain-specific use cases [10]. While this approach facilitates transfer learning across datasets, it often leads to overfitting, which limits the model's generalizability [11]. Layer freezing helps address this issue by enabling selective fine-tuning of specific layers, allowing the model to specialize while preserving general knowledge [12].

2.2. Cross-Dataset Adaptation via Layer Freezing

One of the ways to incrementally enrich the knowledge of models across multiple datasets is through continuous learning [13]. However, this process suffers from the problem of catastrophic forgetting [14]. Under this phenomenon, models tend to lose information that was acquired during pre-training when they are fine-tuned on an unseen dataset. One of the proposed methods to mitigate this issue is layer freezing [15], which is a regularization technique to allow fine-tuning only on specific layers of the model [12]. In our approach, we use selective layer freezing strategy to prevent overfitting during domain adaptation across the BC5CDR, MACCROBAT and Chia datasets. In particular, we freeze the bottom 8 out of 12 transformer layers to selectively fine-tune the top 4 layers. This approach ensures that the general linguistic features are preserved in the lower layers while the higher layers can specialize in learning the patterns that are specific in the new domain.

3. Dataset and Preprocessing

3.1. Dataset for Fine-tuning

To enrich our model with the knowledge of different subdomains of biomedical resources, we use scientific documents, patient reports, and clinical trials. We select one dataset from each of these subdomains for the process of continuous learning in our NER model. Specifically, we select the BC5CDR dataset based on scientific documents for disease and chemical entity recognition, the MACCROBAT dataset for patient reports, and the Chia dataset for clinical trials. The selected datasets represent

different resources and aspects of biomedical NLP which are then utilized for continuous learning approach for model fine-tuning in NER.

3.1.1. BC5CDR

The BioCreative V Chemical Disease Relation (CDR) or BC5CDR corpus [5] provides 1,500 Pubmed documents consisting of 4409 annotated chemicals, 5818 diseases, and 3116 chemical-disease interactions.

3.1.2. MACCROBAT

The MACCROBAT dataset [16] provides an extensive dataset of annotated patient medical reports. As a result, it creates a strong basis for working with and developing clinical NER methods. It is comprised of 200 documents where each of them is annotated with event, entity-types, and the relations. There are 4541 sentences and 98,038 tokens in total, with 22.7 sentences on average per document. The dataset also addresses the challenge of adaptability by covering different concept types in medical information which are not constrained by a specific type of controlled vocabulary or coding system.

3.1.3. Chia

The Chia dataset [6] is an annotated corpus of patient eligibility criteria from patient clinical trials that are registered in ClinicalTrials.gov[2]. There are in total 12,409 annotated eligibility criteria, further containing 15 types of entities and 12 relation types. The entity types in this dataset are further categorized into Domain, Field and Construct, out of which Domain is a mandatory component and the rest are optional. Domain comprises of further eight categories, which are observation, condition, person, device, drug, visit, procedure and measurement. For our use-case, we use condition, drug and procedure as relevant entity-labels in information extraction.

During the pre-processing stage, the passage texts from eligiblity criteria are concatenated on the document level and using BioBERT's tokenizer, tokenized into sequences of maximum 128 tokens. The relevant entity-labels from the Domain category are mapped to the tokens based on character offsets, thereby ensuring a standardized conformity between annotations and the subword units. With its different annotation conventions, the Chia dataset therefore, serves as a cross-domain evaluation dataset to test the robustness of our model.

The three datasets considered in this study (BC5CDR, MACCROBAT, and Chia) follow different annotation guidelines and label granularities. To enable joint fine-tuning, all datasets were converted to a unified BIO tagging scheme at the token level. Each dataset was first processed independently using its original entity annotations, which were then transformed into BIO-formatted labels corresponding to dataset-specific entity types.

4. Methodology

4.1. Model Architecture and Training Strategy

We make use of a transformer-based token classification architecture, namely the BioBERT model which has already been pre-trained on the PubMed and PMC biomedical corpora. This model employs a classification head for predicting the BIO-tagged entity labels. After our model is fine-tuned on the BC5CDR dataset, the weights are then transferred to a new training pipeline for MACCROBAT, followed by the Chia dataset.

A key challenge with this approach is the problem of overfitting, for which a layer-freezing method is used. In this method, we freeze all the encoder layers except the final layer consisting of the classification head. Subsequently, only this classification head is fine-tuned along with a learning rate of $1e-5$ and a batch size of 16 to ensure stability of the model across multiple resources. The overall goal of this

[2] https://clinicaltrials.gov/

approach is to make sure that a general biomedical contextual understanding is retained, while adapting to smaller and nuanced resources.

4.2. Mapping Labels to Controlled Vocabulary

Working with different clinical datasets, each with its own distinct set of labels and categories, introduces lack of uniformity and homogeneity. To standardize the labels, we designed a controlled vocabulary with a fixed set of labels that are relevant to the scope of our use-case. We use the FHIR[3] standards to define four main classes:

Condition[4] To identify the disease, symptoms and other conditions within patients. We also define sub-classes of Condition comprising of Diagnosis Code (MeSH ID), Affected Region, History and Associated Symptoms.

Patient[5] Provides information about the subject under diagnosis or treatment. We use this resource to also provide information about the participants of clinical-trials. The information comprises of "Age" and "Gender" labels.

Medication[6] This class is primarily used for medication that is prescribed and administered, including the constituents such as chemicals.

Procedure[7] Used for treatments that are performed on the patients. This comprises of surgery, therapy and involves both physical intervention as well as less invasive procedures.

The normalization takes place once the entities and their labels are recognized through NER, after which they are mapped to the four predefined FHIR resource classes in our controlled vocabulary. The mapping is performed through a rule-based normalization that aligns entity labels to the controlled vocabulary. Each recognized entity is aligned with one FHIR resource type, based on its label, while sub-attributes such as diagnosis codes, affected regions, or patient demographics are represented via structured fields present within their respective resources.

Given the query “A 65-year-old female patient was diagnosed with breast carcinoma and treated with tamoxifen,” the NER system identifies the entities “65-year-old” (Age), “female” (Gender), “breast carcinoma” (Disease), and “tamoxifen” (Medication).

During post-processing, “breast carcinoma” is normalized and mapped to the FHIR Condition resource, and its corresponding MeSH identifier is represented via a diagnosis code. The demographic entities are mapped to the Patient resource, while “tamoxifen” is assigned to the Medication resource. By linking the extracted entities to FHIR resources, interoperability is ensured across multiple, heterogeneous clinical datasets. Furthermore, the representation of entities through healthcare standards results in structured outputs that can be integrated with Electronic Health Records (EHR), clinical trial databases, and decision-support systems. Our usage of controlled vocabulary also ensures independence from data-specific annotation schemes, improving reusability across multiple sub-domains of clinical tasks.

5. Experiment and Results

We adopted BioBERT as base encoder for our task due to its ability to capture contextual relationships in the clinical dataset. We carried our experiments in two steps: In the first experiment, we fine-tuned the BioBERT model using individual datasets, followed by the second experiment where sequential fine-tuning on the combination of datasets through layer freezing is performed. For evaluation, we used the standard token-level metrics that are commonly used in named entity recognition: precision, recall

[3] https://www.hl7.org/fhir/
[4] https://build.fhir.org/condition.html
[5] https://build.fhir.org/patient.html
[6] https://www.hl7.org/fhir/medication.html
[7] https://build.fhir.org/procedure.html

and F1-score. To address class imbalances and ensure consistency across all the datasets, we computed the metrics using weighted averaging scheme in the scope of BIO format tagging.

5.1. Experiment I

In a first experiment, BioBERT was fine-tuned on each of the three datasets such as BC5CDR, MACCROBAT, and Chia. On each of these three datasets, training was performed for 5 epochs using the AdamW optimizer [17]. In this first experiment, no layers of the model were frozen such that both the encoder and classification heads were trained and updated jointly. Results of the evaluation are given in Table 1.

Table 1
Performance of BioBERT when trained on individual datasets. The results are stated in percentage.

Model	Precision	Recall	F1
BioBERT + BC5CDR	79.8	90.0	84.6
BioBERT + Chia	59.42	52.76	55.4
BioBERT + MACCROBAT	54.27	70.6	61.3

In this experiment, we observed BC5CDR to have performed the best. We attribute this largely balanced nature of the dataset, further supplemented by an abundance of training examples.

Chia remains to be a challenging dataset for model training and fine-tuning, owing to fewer training examples, lexical diversity and highly variable span lengths, making it difficult for token-level BIO NER. The relatively smaller dataset of MACCROBAT also affects the performance, resulting in a low F-score, owing to its reduced precision.

5.2. Experiment II

In the second experiment, we used continuous learning to enrich BioBERT with the ability to generalize across all three of the datasets. We used BC5CDR as our initial dataset, and by using sequential fine-tuning and selective freezing, proceeded with MACCROBAT and Chia datasets, separately and together to enrich the base model. The evaluation results obtained are shown in Table 2.

Table 2
Performance of BioBERT when fine-tuned on joint datasets by layer freezing.

Model	Precision	Recall	F1
BioBERT + BC5CDR + Chia	71.52	71.70	71.26
BioBERT + BC5CDR + MACCROBAT	84.00	83.00	82.70
BioBERT + BC5CDR + Chia + MACCROBAT	86.00	85.40	85.00

From the results obtained, the sequential fine-tuning by layer freezing resulted in an improvement over individual fine-tuning of the datasets. While BC5CDR provided an extensive foundation of annotated PubMed documents, its scope is limited to chemical and disease labels. However, a model enriched with the knowledge from BC5CDR provided an improvement in performance during fine-tuning on MACCROBAT, more than what was achieved with BC5CDR and Chia.

By further fine-tuning on the combination of all the three datasets, the model learned from additional resources and was able to scale without catastrophic forgetting and overfitting. In the end, we used the labels from MACCROBAT dataset, as they fall within the scope of our current use-case.

For a better illustration of a potential clinical impact of our approach and these results, consider a patient discharge summary describing medical history with multiple comorbidities, medications, and ongoing procedures. A model trained solely on BC5CDR may reliably recognize disease and chemical mentions but would not be able to capture other fine-grained contextual entities commonly found in clinical texts. Therefore, by enriching the base model through joint fine-tuning with BC5CDR, Chia,

and MACCROBAT to achieve higher precision and recall, we will improve its ability to capture those contextualized entities in patient profiles, eligibility screening for clinical trials, and clinical decision support to provide a comprehensive patient representation.

6. Conclusion

This paper tackles the fundamental problem of dataset unavailability in the biomedical domain, specifically through continuous learning. Through sequential fine-tuning and layer freezing, our approach improves BioBERT's performance across three diverse biomedical domains. We adapt the model across complex and challenging datasets, which, when sequentially fine-tuned, provide significantly better results than their individual fine-tuning. The combined model (BioBERT + BC5CDR + Chia + MACCROBAT) demonstrates knowledge transfer across datasets without the added penalty of overfitting and catastrophic forgetting. We aim to integrate this model into a user interface that can extract relevant clinical concepts across inputs from domains such as scientific publications, clinical trials, and patient profiles. For our future work, we wish to explore and evaluate additional domain adaptation techniques and also use synthetic biomedical data to enrich smaller datasets for further specialized NER tasks.

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Declaration on Generative AI

During the preparation of this work, the authors used Mistral AI for grammar and spelling check. After using this service, the authors reviewed and edited the content as needed and take full responsibility for the publication's content.

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